

NWFP Group Management and Marketing Plan



Name of NWFP Group: Ngangla-Trong Shingmen Tshogpa

Village: Ngangla and Trong

Gewog: Ngangla

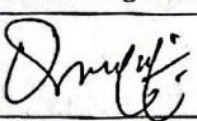
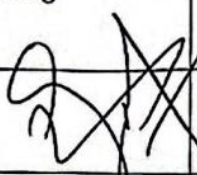
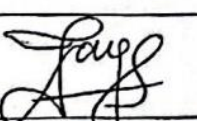
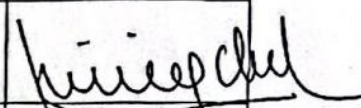
Dzongkhag: Zhemgang

Plan Period: 2024 –2034 (*1st Revision*)

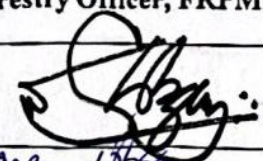
Approval Sheet

Name of NWFP Management and Marketing Group	Ngangla-Trong Shingmen Tshogpa
Village(s)	Ngangla, Trong
Gewog	Ngangla
Dzongkha/Dungkhag	Zhemgang
No. of households	41
Area (ha)	32
Plan period	2024-2034

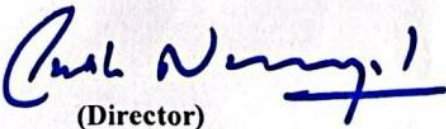
Prepared and submitted for Approval

	Chairperson of NWFP Management and Marketing Group	Gup of the Gewog	Head FRPMS of Division/Park	Chief Forestry Officer of Division/Park
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Date:				

Reviewed and recommended for Approval

	Reviewing Officer, FRPMD	Chief Forestry Officer, FRPMD
Signature:		
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Date:	25-26-2024	25/07/24

Approved By


(Director)

Department of Forests and Park Service, MoENR

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Acronyms

AHL	Annual Harvesting Limit
BDBL	Bhutan Development Bank Limited
DFO	Divisional Forest Office
hh	Household
ITMS	Institute of Traditional Medicine Services
Kg	Kilogram
Km	Kilometer
ac	Acre
ha	Hectare
KNC	Khenrig Namsum Cooperatives
NWFP	Non-wood Forest Produce
FRPMD	Forest Resources Planning and Management Division
FRPMS	Forest Resources Planning and Management Section

Part A: NWFP Management and Marketing Plan

1. Background

Ngangla Trong Non-wood Forest Product (NWFP) Management Group is located under Ngangla Gewog, Zhemgang Dzongkhag. It is about one hour drive from Ngangla Gewog Center at Panbang. The community has access to modern amenities like RNR-Extension service, forestry service, electricity, water supply, motorable road and telecommunication network. Agriculture is primary source of livelihood in the village and people cultivate maize, rice, potatoes, beans, cabbage, chili and cash crops like orange, avocado cardamom and ginger. They also depend on forest for timber, firewood and NWFPs for additional income generation for their livelihood. The village has 41 households and all are registered member of Ngangla Trong Shingmen Tshogpa. There are various NWFPs found in their locality such as *Neomicrocalamus andropogonifolius* (Yula), *Diplazium esculentum* (Jeema), *Knema tenuinervia* (Durka), *Borinda grossa* (Baa), *Piper betleoides* (Dhem pan), *Oroxylum indicum* (Namkaling), *Thysanolaena maxima* (Tsakusha), *Musa* and its part (*Mochali* and *pachali*), *Tupistra nutans* (*Ngawa pem/Thorma Ju*), *Cane* spp. (*Khran khengkha*), *Dendrocalamus hamiltonii*, mushroom and wild yam. Though community harvests all those species for self-consumption & few for commercialization, they have prioritized *Dendrocalamus hamiltonii*, *Neomicrocalamus andropogonifolius* (Yula) wild yam, *Tupistra nutans* and mushroom for commercialization. There are four bamboo management areas consisting 32ha which will be management by community during the plan period.

During the previous management plan, community couldn't implement the activities as per plan due to Covid-19 pandemic. However, the group has marketed the bamboo shoot to Khengrig Namsum Cooperatives (KNC). The previous management plan had expired in June, 2022. Before reviewing the management plan, a comprehensive consultation meeting with the management group members was conducted to understand their inactiveness and to assess their interest in continuing further management. Taking their interest in continued management of NWFPs, Divisional Forest Office (DFO) Zhemgang provided technical support to the group in resource assessment and preparation of management plan. The management plan will guide the group members to implement the activities in order to ensure the sustainable harvesting of NWFPs.

2. Goal and objectives

1. To achieve self-sufficiency through management and marketing of NWFPs products in sustainable basis
2. To up lift the life of poor people in the rural village through marketing the NWFPs products
3. To generate income for the local communities.

3. Description of NWFPs present in the area demarcated for management

Dendrocalamus hamiltonii: It is a bamboo species with straight strong (near solid) culms and edible shoots. It has very tight clumping growth habit and slightly weeping tops with lush foliage. Some of the culms may have a cream striation on the lower inter-nodes. Culms is 9-25cm in diameter, strong pendulous above, densely covered in persistent, brown and white, furry and wax; wall thin; branches fewer toward base, center branch to 5cm in diameter, smallest branches curving from clump, nodes with dense, long aerial roots. The group will collect the resources from Bangsekhar, Pongchalking, Choekhar and Umchomi blocks. It is around one hour drive from the village.

Neomicrocalamus andropogonifolius: It is a perennial plant with the following characteristics: pluricaespitose, elongated rhizomes, scandent culms, it grows up to 1200cm long, woody, culm-internodes terete, thin-walled, culm nodes flush with internodes, leaf-blade base broadly

rounded, leaf-blade margins scabrous and leaf-blade apex acuminate. It is mainly used for crafting purposes. Management group wanted to supply the resources to Bjoka NWFP management group for bamboo crafting and also wanted to train themselves for crafting to bamboo production. But due to flowering of yula, assessment couldn't conduct during the resource assessment for revision of this management plan.

Diplazium esculentum: It is a straggling fern that grows up to a meter tall. The young shoots or fronds of the fern are consumed as vegetables, sold in local markets throughout Asia and the Pacific Islands. Fronds are variable in size, bipinnate, linear leaflets with lobed to serrated margins. Is the vegetable fern, edible fern found throughout Asia and Oceania. It is probably the most commonly consumed fern. The management group would collect the resources within their managed area and wanted to sell fresh or processed fern in the market.

Knema tenuinervia: Is an evergreen shrub or tree, usually growing from 3 - 20 meters tall with some specimens reported to reach 30 meters. The tree is harvested for its wood, which is used locally and especially for processing into traditional medicines.

Mushroom: Edible mushrooms are macrofungi since they are visible with a naked eye. Mushrooms have fleshy fruit body with certain aroma and flavors as well as good nutritional properties and are used mostly as food. The community wanted to collect those edible mushroom and market for their income generation.

Piper betleoides: The betel plant is an evergreen perennial, with glossy heart-shape leaves and white catkin. The betel leaf plant is a creeper grown in our country too and belongs to pepper family. Its waxy green, heart-shaped leaves are used for medicinal and culinary purposes. Community prefers the resources to collect from the vicinity of managed area for marketing.

Oroxylum indicum: Is a monotypic species of flowering plant belonging to the genus *Oroxylum* and the family Bignoniaceae, commonly called midnight horror, *oroxylum*, or Indian trumpet flower. It is a mesocaul tree which can reach a height of 18 meters. The resources are also presence within the managed NWFP area of the group.

Thysanolaenna latifolia: A strongly tufted, very robust perennial grass with erect or slightly spreading solid bamboo-like culms up to 3.5 m tall. Leaf-sheath hairy along outer margin; ligule a scarious membrane, 1-2 mm long; leaf-blade lanceolate-acuminate, 30-65 cm x 3-7.5 cm, base broad and rounded or subcordate, margins scaberulous, conspicuously glaucous beneath. Inflorescence a terminal huge and drooping panicle, up to 140 cm long, well exerted, branches divided and subdivided into many branchlets; spikelets awnless, short pedicelled, falling with part of the pedicel, often in pairs on a common peduncle, 2-flowered; lower glumes clasping. Caryopsis subglobose to ovoid, 0.6 mm long, reddish-brown. Members wanted to collect the resources markets after processing.

Tupistra nutans: is an amazing Indian native that makes a nice clump resembling an aspidistra on steroids. Each clump is composed of 3' long narrow dark green leaves that are held upright. The flower of species is used as vegetable which is sold expensively in the market.

Plectocomia himalayana: Stems 3-15m or more, scandent. Leaf sheaths tomentose (orange-brown in lower part, pale fawn above), heavily armed with oblique or wavy-horizontal lamellae bearing unequal (longest to 2.5cm), slender, straight spines; mouth oblique, auricles not developed. Petiole very short, not kneed. Rachis flattened; fawn-tomentose, lower part sometimes spiny, upper part developed into long cirrus (to 2m) heavily armed with grapnels. Leaves to 3.5m, pinnae relatively few (to 25), linear-lanceolate, gradually drawn into long (to 8cm) filiform apex, longest 30-45cm, widest 2-3.6cm, irregularly inserted-singly or in groups of 2-3, margins, spinulose, with many, glabrous, parallel veins.

Wild yam: Wild yam is perennial climbing vine that is edible. But there are some of the yams which cannot be eaten. Community will collect those edible yams for marketing.

3.1 Brief description of NWFPs for commercial harvesting in each season

NWFP name	Harvesting area(s) (name)	Parts harvested	Use/purpose (traditional & others)	Harvesting season (months)	Estimated quantity for harvest in a season/year
<i>Neomicrocalamus andropogonifolius</i>	Choekar, Objang, Pongchaling	Culms	Crafting	December-May	900 bundles
<i>Dendrocalamus hamiltonii</i>	Choekar, Objang taa, Pongchaling, Garbaling, Nganglatrong	Shoot	Vegetables	May-September	5000kg
		Culm	Timber	12 months	3000 pieces
<i>Elastostema lienolatum</i> (Damroo)	Choekar, Objang taa, Pongchaling, Garbaling, Nganglatrong	Young shoot & leave	Vegetables	April-August	1500 bundles
<i>Knema tenuinervia</i>	Choekar, Objang, Pongchaling	Woods	For religious	Whole year	1500kgs
Wild yam	Choekar, Objang taa, Pongchaling, Garbaling, Nganglatrong, Umchomi	Tuber	Food	December - May	1000kg
<i>Piper betleoids</i>	Choekar, Objang taa, Pongchaling, Garbaling, Nganglatrong	Leaf	Stimulant	Throughout the years	1500 bundles
<i>Oroxylum indicum</i>	Choekar, Objang taa, Pongchaling, Garbaling, Nganglatrong, Umchomi	Flower	Vegetables/d yestuff/religious	December to March	1000pieces
<i>Thysonolaenna latifolia</i>	Choekar, Objang taa, Pongchaling, Garbaling, Nganglatrong, Umchomi	Upper parts, inflorescence	Fodder/soft broom	December to March	2000 bundles
Mushroom	Choekar, Objang taa, Pongchaling, Garbaling, Nganglatrong, Umchomi	Whole parts	Vegetables	May-Oct	2500 kg
<i>Tupistra nutans</i>	Choekar, Objang taa, Pongchaling, Garbaling, Nganglatrong, Umchomi	Flower	Vegetables	Sept-Dec	1000kg
<i>Plectocomia himalayana</i>	Dimphu	Shoot/Ste m	Crafting	Dec-May	2500 pieces

4. Procedures for sustainable harvesting of NWFPs

The NWFP management group members have to adhere to specific procedure for the sustainable management of resources.

- The Chairperson will consult with forest office and obtain permit for collection during the onset of harvesting season.
- The Chairperson will announce starting date for harvesting only after obtaining required permits from forest office.
- Only one member from each household will be entitled for collection of NWFP products.
- Collection will be done within the demarcated area only.
- Harvesting prescriptions will be strictly followed by the collectors.
- The Vice-Chairperson will maintain proper record of harvest and sale of NWFPs for monitoring and evaluation.
- Chairperson will notify the forest office on completion of harvesting the NWFPs.
- Treasurer of NWFPs management group to maintain all financial records

Table 1. Prescriptions for sustainable harvesting of NWFPs

NWFP Species	Harvesting prescription
<i>Neomicrocalamus andropogonifolius</i>	▪ Cut only feasible culm ▪ Do not cut all culm & retain atleast two - Harvest culm when it produces secondary shoot
<i>Dendrocalamus hamiltonii</i>	• Collect shoot up to 25% from the clump • Use sharp knife to cut the shoot without damaging other parts in the stand • Cover the cut portion of the shoot that is left in the stand with soil
<i>Diplazium esculentum</i> (fern)	❖ Collect the fern that is above the ground level only ❖ Do not up root the whole plants ❖ Just break the base of the fern to collect the shoot
<i>Knema tenuinervia</i>	✓ Collect only fallen trees ✓ Never cut standing trees
<i>Borinda grossa</i>	➤ Collect matured culm ➤ Do not disturb the young shoot ➤ Harvest the culm from the base to maintain sanitation
<i>Piper betleoides</i>	❖ Collect green leave ❖ Do not cut down the trees ❖ Do not break down the stem ❖ Climb up the trees to collect the leaves seeing the possibilities
<i>Oroxylum indicum</i>	• Collect flower when matured • Never cut down the trees • Handle with care while harvesting the flower
<i>Thysonolaenna latifolia</i>	➤ Collect the matured parts/stem ➤ Never cut it from the base ➤ Do not uproot the stem
<i>Wild Yam</i>	○ Collect just matured tuber ○ Do not collect all tuber from same area ○ Re-cover the pit after digging yam ○ Harvest sustainably, it is easier to vanish
<i>Tupistra nutans</i>	❖ Just collect flower only ❖ Never uproot the stem
<i>Plectcomia himalayana</i>	➤ Cut only matured cane ➤ Cut canes with only 10m usable length ➤ Collect young cane shoot by bending - Do not cut the tree that support the cane
Mushroom	➤ Do not fell down the trees ➤ Collect the mushroom from dried trees ➤ Do not harm the host trees while collecting the mushroom

4.1 Block-wise area and distances for prioritized species

There are various collection areas for different NWFPs that will manage by Ngangla-Trong Shingmen Tshogpa but very minimal. The areas and walking distances for prioritized species especially for *Dendrocalamus hamiltonii*, Yula, Mushroom and Wild yam is as tabulated below.

Table 2. Block-wise area detail

Village	Collection area	Distance from village	Total area(ha)
Ngangla Trong	Umchomi	1hr	10.44
	Choekar	0.40 hrs	6.25
	Brangseekhar	0.40 hrs	5.4
	Pongchaling	1.30 hrs	9.91
Total			32ha (79.07acres)



Figure 1. Map of NWFP management area

5. Marketing and sales plan

NWFPs collected by Ngangla Trong Shingmen Tshogpa plans to market and sell NWFP to listed potential markets:

1. National Institute of Traditional Medicine Service (NITMS)
2. Food Cooperation of Bhutan (FCB) which will be operated as farmers shop at geog level
3. intruder who deals with the resources
4. Khengrig Namsum Cooperative at Tingtibi, Zhemgang
5. Local Markets at Panbang
6. Nearby border towns
7. Thimphu CSO market
8. Yula can be sold to Bjoka Tsharzo Gongphel Tshogpa/ and individual people of Bjoka community

The quantity, price of products & the value of expected sales will be done as per the data listed in the table 2 below. Yula resources assessment couldn’t conduct due to mass flowering during resource assessment time.

Table 3. Estimated quantity and sales plan

NWFP	Potential Market	Estimated quantity for sale per season (unit)	Expected price per unit	Total (Nu)
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<i>Neomicrocalamus andropogonifolius</i>	Bjoka Tsharzo Gongphel Tshogpa & Bjoka people individually	900 bundles	400/bundle/mtr	360000/=
<i>Dendrocalamus hamiltonii</i>	Khengrig Namsum Cooperative at Tingtibi, Zhemgang	5000kg (Shoot)	12/kg	60000/-
		3000 pieces (Culm)	20/piece	60000/-
Damroo (<i>Elastostema lienolatum</i>)	Local market	200 bundles	15/bundle	3000/-
<i>Knema tenuinervia</i> (Durka)	Local Markets at Panbang	1500kgs	30/=	45000/=
Wild yam	ITMS, border towns and intruder	1000kg	100/=	100000/=
<i>Piper betleoides</i>	Border towns, intruder and Local Markets at Panbang	1500 bundles	5/=	22500/=
<i>Oroxylum indicum</i>	Border towns, intruder and Local Markets at Panbang	1000pieces	10/=	25000/=
<i>Thysonolaenna latifolia</i>	Border towns, intruder and Local Markets at Panbang	2000 bundles	20/=	40000/=
Mushroom	Borderr towns, intruder and Local Markets at Panbang	1000kg	150/=	150000/=
<i>Tupista nutan</i>	Flower	100kg	150/kg	15000/-
<i>Plectocomia himalayana</i>	Stem	2000 pieces	35/piece	70000/-
Total Amount projected				950500/-

5.1 Marketing & Sales plan for prioritized bamboo products

The community of Marangduth NWFPs group has also projected the sales plan for prioritized *Dendrocalamus hamiltonii* resources mentioned underneath in the table 4.

Table 4. Sales plan for prioritized species

NWFP	Products (Local Name)	Unit	Rate per kg/Piece
<i>Dendrocalamus hamiltonii</i>	Tangloop	Piece	450/-piece
	Basket	piece	800/-piece
	Bamboo hat	piece	500/-piece
	Roong	piece	1000/piece
	Zortay	piece	100/-piece
	Shera	piece	350/-piece
Mushroom	Moo	kg	150/kg
Wild yam	Keey	kg	100/kg

6. Forest Resource Assessment data for *Dendrocalamus hamiltonii*

Table 5. Resources assessment data

	All Block mentioned		Dates of bamboo resource assessment:	27-29/1/2024		
Block name:						

Sub-block (number)	Name of sub-block		Area (ha)	% of productive area		Number of sample plots
Umchomi	Umchomi		10.44	97		13
Choekar	Choekar		6.25	95		9
Brangseekhar	Brangseekhar		5.4	94		8
Pongchaling	Pongchaling		9.91	95		12

Age class	Number of culms	Block 1(Umchomi)	Block 2(Choekar)	Block 3(Brangseekhar)	Block 4(Pongchaling)	Total in block
1+ year old bamboo culms	Total counted in sample plots	96	62	53	44	
	Total harvestable in sample plots	61	55	48	34	198
	Total in block*	3840	2480	2120	1760	10200
	Total Harvestable in block*	2440	2200	1920	1360	7920
2+years old bamboo culms	Total counted in sample plots	81	54	52	45	232
	Total harvestable in sample plots	71	48	46	28	193
	Total in block*	3240	2160	2080	1800	9280
	Total Harvestable in block*	2840	1920	1840	1120	7720
3+years old bamboo culms	Total counted in sample plots	98	62	56	49	265
	Total harvestable in sample plots	87	54	43	34	218
	Total in block*	3920	2480	2240	1960	10600
	Total Harvestable in block*	3480	2160	1720	1360	8720
Total		20254	13735	12218	9360	46207
AHL		5080	3266	2953	4310	11301

AHL= 33% of the total stock of the 1+year old culms +2+year old culms+ 70% of 3+years old culms

In which the 33% of the stock of the 1+year old and the 2+years old culms should not exceed the total of the harvestable culms of the combined age classes.

* As calculated per productive area in sub-block/block(calculate the average per sample plot and divide by 0,025(10*25)/10000)to get the number of culms

per ha,multiply with the productive area(ha) of sub--block/block to get the total of culms per

sub-block/block: compare the result with the culms considered harvestable by the villagers (calcuated in the same way)

6.1 Demand & annual harvesting limit for prioritized Bamboo products

Table below denotes the demand of community members for *Dendrocalamus hamiltonii* shoot and culms and its annual harvesting limits.

Table 6. Annual demand and AHL

Total Annual Demand	Annual Harvesting Limit	Variation	Remarks
3000 pieces culms	11301 pieces of culms	9431 pieces of culms	Surplus

7. Seasonal calendar for NWFP management group

The group members shall abide by the seasonal calendar provided in table.....for implementation of activities.

Table 7. Seasonal calendar for activities

Program	Bhutanese Calendar											
	1	2	3	4	5	6	7	8	9	10	11	12
Collection of Bamboo shoots and processing												
<i>Neomicrocalamus andropogonifolius</i> collection and trading												
<i>Diplazium esculentum</i> (fern) collection and marketing												
<i>Knema tenuinervia</i> collection and marketing												
<i>Borinda grossa</i> collection and marketing												
<i>Piper betleoides</i> collection and marketing												
<i>Oroxylum indicum</i> collection and marketing												
Plantation activities for selected species												
Weeding and cleaning activities for all species												
Collection and marketing of <i>Thysonolaenna latifolia</i>												
Collection and marketing of <i>Knemia</i> spp.												

8. Marketing Arrangement.

Following were the marketing strategies planned for the Ngangla Trong Shingmen Tshogpa during the group formulation.

1. Products will be collected & transported up to the road point by the members
2. Products like cane shoot, mushroom, damroo, fern etc. will be sold in local market
3. Piper will be sold to nearby Indian town and other intruders
4. Yula to local markets & finished products at Thimphu
5. Cane to Bjoka Tsharzo Gongphel Tshogpa and other traders.
6. *Dendrocalmus hamiltonii* to KNC, Thimphu and other market areas.

8.1 Procedures for NWFP harvesting, marketing and sharing of income

Activity	Brief description of how the activities are carried out
Harvesting of products	• The Chairperson of NWFP group will obtain harvesting/collection permit from forest office in writing. • The Chairperson will announce starting date for harvesting only after obtaining required permits from forest office. • Only one member from each household will be collect NWFP for the first week and after that no limits is set for member involvement for collection. • Harvesting prescriptions will be strictly followed by the collectors. • Chairperson shall inform Range Office of completion of harvest.
Postharvest practices	• The collectors are responsible for post-processing of NWFPs.

Marketing of products	• The NWFPs will be handed over to the executive members for marketing after post-processing. • The executive members will sell the produce in attendance of the group members to maintain transparency. In future the group plans to construct a common sale counter at the center to facilitate marketing of the NWFPs products.
Identification of buyers	• The Chairperson supported by other executive members will monitor and explore the demand from nearby institutes, agents, companies and other business groups. • Support will also be sought from Department of Agriculture and Marketing Cooperatives (DAMC)
Price fixing	• The executive members will coordinate.
Income sharing	• . The income generated from the sale of NWFPs will directly share among the members. All the members will have equal right for sustainable harvesting of products as per the entitled lists in the plan. The group will also utilized the common fund during the community rituals and natural calamities with basic understanding among the members
Raising of community development fund	• A joint account in the name of the group will be opened under BDBL Branch Office located at Panbang. The account shall be managed by the committee members on behalf of the group. • Annual membership fees, penalties and certain amount of sale proceeds will be deposited in the community development fund.
Payment to committee members for their service	• 5% of total income from the sale of products will be deposited into group's account & 15% from the sale will be shared within the committee workers as work charges.

9. Benefit to the members

The income generated from the sale of NWFPs will directly share among the members. All the members will have equal right for sustainable harvesting of products as per the entitled lists in the plan. The group will also utilize the common fund during the community rituals and natural calamities with basic understanding among the members.

Table 8. Projected revenue from sales of NWFP

Sl. No	Particulars	Products (names)				Total Amount (Nu)
		Dendrocalamus hamiltonii		Mushroom	Wild yam	
		Shoot	Culm			
1	Revenues					
A	Total quantity (unit)	5000kg	3000piece	1000kg	1000kg	
B	Market price (Nu per unit)	12	20	150	100	
C	Total revenues from forest product sales (a*b)	60000	60000	150000	100000	370000
A	Total revenues					370000
2	Expenses					
A	Packaging materials	-		-	-	-
B	Royalty	5000 pieces@12			-	60000
C	Transportation costs					30000
D	Compensation fee to executive committee for marketing products	-		-	-	-

E	Contribution to community development fund	5% of total revenue	-	-	18500
F	Incidental expenses	-	-	-	-
B	Total expenses				108500
C	Total benefit (net-income) of group members (Total revenues A – Total expenses B)				261500
D	Average income to group members (C/ no of group members)				7067.57
E	Price per unit paid to group members (C/1a)				26.15

Table 9. Benefit projection to CDF

Sl. No	Particulars	Amount (Nu)
	Revenues	
1	Membership registration fees (37hh @100)	3700
2	Annual contribution (5% from total revenue)	18500
A	Total revenues	22200
	Expenses (only include expenses that will be paid from the group funds)	-
		
B	Total Expenses	-
C	Balance: Total revenues – Total expenses	22200

Part B: Group Regulation and Norms (By-Law)

NWFP management & marketing group	
Membership arrangement and resignation	Any household with census recorded in Ngangla Trong village and residing in the area can join the group. However, acceptance in the group will be made only by the group members through assembly & submitting letter of interest to the committee members. <u>Responsibilities and rights of members:</u> • Implement annual activities that are reflected in the management plan. • Manage the area in adherence to the management plan and refrain from doing illegal activities. • Attend ad-hoc/ planned meetings or activities for development of the group. • Attend any training related to NWFP. • Choose or reject committee members for better management of the group. <u>Procedure for becoming new members:</u> • Any household wishing to become members in the group shall submit application to the Chairperson. • The applicant should be 18 years of age to become a member. • The application shall be put forward in the general meeting for discussion and every member has right to accept or reject the application. • Membership fee of Nu.1000/- shall be levied upon registration. <u>Procedure for leaving the group:</u> • Any member wishing to discontinue their membership has right to do so. However, he/she should submit application to the Chairperson. • He/she can leave the group after discussing with the committee members. • Member leaving the group will not be allowed to avail their benefits from the group fund. • If their children wish to join as a member in future, they have to follow the procedure set to become new member.
Executive committee of NWFP management and marketing group	
Chairperson	Name of Chairperson: Leki Drakpa Contact No. <u>Responsibilities of Chairperson:</u> • Chair all the group meetings in which the group will have half yearly meeting twice in a year & a general meeting • Explore market for NWFPs. • Obtain collection permit from forest office during the onset of harvesting season. • Coordinate and plan activities for the user group along with the members and other stakeholders. • Coordinate and conduct meetings with members for development of the group. • Resolve conflict among group members. • Submit annual progress report to forest office. • Maintain record of all the group activities.
Secretary	Name of Secretary: Pema Zangpo Contact No. <u>Responsibilities of Secretary:</u>

	• Officiate in absence of Chairperson. • Assist Chairperson in coordinating and planning group activities. • Explore market for NWFPs. • Maintain records of any activities conducted by the group. • Maintain attendance record while conducting meeting and group activities. • Record minutes of meeting.
Treasurer	Name of Treasurer: Karma Choeda Contact No. <u>Responsibilities of Treasurer:</u> • Officiate in absence of Chairperson and Secretary. • Collect annual membership fees, fines, amount from sale proceeds and deposit in the group account. • Maintain proper financial records in favor of the group. • Present financial status/ progress report during general meeting. • Assist Chairperson and Secretary in coordinating and planning group activities.
Tenure of committee member	• Their tenure shall be till the plan period ends. • However, if a committee member does not fulfill his/her responsibilities, misbehave and threaten the group's network, he/she shall be immediately replaced and be deprived of such opportunity in future.
Incentives to the committee members	• If the committee members have worked positively towards betterment of group, he/she shall be re-elected. This is to reward the individual for his/her dedicated service to the group.
Resoop	• Conduct Patrolling twice in a year • Report illegal activities to the committee members
Sources for generation of group fund	
Membership registration fees Annual contribution Collection from sale proceeds	• Membership fees of Nu.100 per member household shall be collected annually. • The amount deposited will be used as pre-financing cost for marketing of NWFPs products • Amount realized as penalties shall be deposited in the group fund. • 5% of total products sold will be deposited to group account
Group fund management and utilization	• A joint account in favor of the group shall be opened under BDBL, Panbang. • Annual membership fees, fines and amount from sale proceeds shall be deposit in the group account. • The joint account will be managed by Executive Committee members. • Any expenditure incurred in the name of the group will be met from community development fund. • When the fund accumulates more than two hundred thousand, the fund will be utilized for community rituals, supporting the poor and disaster affected household. • If the group gets dissolved under unforeseen circumstances, the entire fund accumulated shall be equally divided among the members. • Welfare scheme will be decided by community member upon open discussion with the member

Conflict resolution	- Disputes between the members will be resolve by committee if not, - The conflicts that arise amongst the group members about their group activities and management areas of NWFP, it will be put up-to Geog Tshogdu (GT) followed by Dzongkhag Tshogdu (DT) - If the conflict arises between NWFP group and outsider, case will be forwarded to DFO and Geog Administration by the committee
Meeting and purpose	In order to plan and discuss activities for the management of NWFPs, meeting will be conducted as follows; General Assembly: All the members shall attend the meeting to discuss and plan activities before harvesting season; during April month every year. Committee member: This meeting is mainly to discuss the planned activities as well as prepare progress reports (financial and physical) during the November month.
Penalties	- If the group members' violet the harvesting rules for NWFPs, he/she should pay the fine of Nu. 250/= & is not allowed to harvest the products in next harvesting season - Any group members selling the products to outsider without notice of group will penalize fine of Nu. 1500/= - If outsider collects the products from the group's demarcated areas, they will be fined of Nu. 500/= if their intention is to commercialize for income generation. - If the collection area is being damaged due to cattle grazing, the owner will be fined Nu.150/= per cattle - Member inactive after one year will be excluded from the group. - Income generated in group will be shared with the active member except inactive member will not be entitled.

Powers of the Department

The Department has the right to terminate the NWFP management and marketing group in case of any violation of the management plan prescription or in the event of inability to carry out the activities as per the plan.

The deregistration, cancellation, termination and reinstatement of the NWFP plan shall be guided by Section 261 to 265 of the Forest and Nature Conservation Rules and Regulations, 2023.

Appendix I: Overview of *Dendrocalamus hamiltonii* collection area

Block name	Sub-block	Total area (Ha)		Production area (%)		Production area (Ha)
Umchomi	Umchomi	10.44		97		10.12
Choekar	Choekar	6.25		95		5.94
Brangseekhar	Brangseekhar	5.4		94		5.1
Pongchaling	Pongchaling	9.91		95		9.41
		32				30.57

Appendix II: Block description form for *Dendrocalamus hamiltonii*
Pongchaling Block

Date: 25/1/2024				
Village:	Pongchaling	Name of bamboo area: Dangling		
Block name:	Pongchaling	Total area (ha)	9.91	

Access	What is the walking distance from the nearest village to the block?			
	Less than 3 hours	Within a day	More than a day	
	What is the walking distance from the block to the nearest road?			
	Less than 3 hours	Within a day	More than a day	

Slope	Very steep >100%	Steep 70-100%	Moderate 45-70%	Gentle <45 %
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Forest Description	Dominant and associate species, canopy cover:				
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Sub-block	No.	Density	Area (acres)	Production area (%)	No. of sample plots
	1		24.48	95	11

Bamboo flowering	Sporadic	Gregarious	Last time of mass flowering	None
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Plantation potential	Sub-blocks	Has plantation potential	No potential	Not applicable
	1. Umchomi	Yes		

Current management	Current management practices in entire block:
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Current threats	Threats	Description
	Fire	
	Grazing	
	Illegal exploitation	Illegal collection by others
	Over-exploitation	
	Other:	

Opportunities	Future opportunities for the entire block: High production area if managed sustainably
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Ngangla Trong Block

Village:	Umchomi		Name of bamboo area: Umchomi	
Block name:	Ngangla Trong		Total area (ha): 10.44	

Access	What is the walking distance from the nearest village to the block?		
	Less than 3 hours	Within a day	More than a day
	What is the walking distance from the block to the nearest road?		
	Less than 3 hours	Within a day	More than a day

Slope	Very steep >100%	Steep 70-100%	Moderate 45-70%	Gentle <45%
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Forest Description	Dominant and associate species, canopy cover:
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Sub-block	No.	Density	Area (acres)	Production area (%)	No. of sample plots
	1		25.79	97	12

Bamboo flowering	Sporadic	Gregarious	Last time of mass flowering	None
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Plantation potential	Sub-blocks	Has plantation potential	No potential	Not applicable
	1. Umchomi	Yes		

Current management	Current management practices in entire block:
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Current threats	Threats	Description
	Fire	
	Grazing	
	Illegal exploitation	
	Over-exploitation	Over exploited due to nearby community
	Other:	

Opportunities	Future opportunities for the entire block: High production area if managed sustainably
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Choeckhar Block

Date: 27/1/2024		
Village:	Choeckhar	Name of bamboo area: Choeckhar
Block name:	Choeckhar	Total area (ha) = 6.25

Access	What is the walking distance from the nearest village to the block?		
	Less than 3 hours	Within a day	More than a day
	What is the walking distance from the block to the nearest road?		
	Less than 3 hours	Within a day	More than a day

Slope	Very steep >100%	Steep 70-100%	Moderate 45-70%	Gentle <45%
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Forest Description	Dominant and associate species, canopy cover:
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Sub-block	No.	Density	Area (acres)	Production area (%)	No. of sample plots
	1 Choekar		15.44	95	8

Bamboo flowering	Sporadic	Gregarious	Last time of mass flowering	None
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Plantation potential	Sub-blocks	Has plantation potential	No potential	Not applicable
	1. Choekar			

Current management	Current management practices in entire block: NWFPs managed area
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Current threats	Threats	Description
	Fire	
	Grazing	
	Illegal exploitation	
	Over-exploitation	Over exploited due to nearby community
	Other:	

Opportunities	Future opportunities for the entire block: High production area if managed sustainably
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Bangsekhar Block

Date: 29/1/2024		
Village:	Bangsekhar	Name of bamboo area: Bangsekhar
Block name:	Bangsekhar	Total area (ha) = 5.4

Access	What is the walking distance from the nearest village to the block?						
	Less than 3 hours		Within a day		More than a day		
	What is the walking distance from the block to the nearest road?						
	Less than 3 hours		Within a day		More than a		

					day		
Slope	Very steep >100%	Steep 70-100%	Moderate 45-70%	Gentle <45%			
Forest Description	Dominant and associate species, canopy cover:						
Sub-block	No.	Density	Area (acres)	Production area (%)	No. of sample plots		
	1		13.34	94	8		
Bamboo flowering	Sporadic	Gregarious	Last time of mass flowering	None			
Plantation potential	Sub-blocks	Has plantation potential	No potential	Not applicable			
	1. Bangsekhar						
Current management	Current management practices in entire block: NWFPs managed area						
Current threats	Threats		Description				
	Fire						
	Grazing		Free grazing by cattle				
	Illegal exploitation						
	Over-exploitation						
	Other:						
Opportunities	Future opportunities for the entire block: High production area if managed sustainably						

Appendix III: Resource assessment form for *Dendrocalamus himaltonii*
Umchomi Block

Date: 27/1/2024
Block: Umchomi
Sample plot no: 1
Aspect: SE

Village: Ngangla
Local name of resource assessment area:Umchomi
Slope: Less than 45%
Altitude: 1279m

Recorder: Sherab Dorji & Ugyen Dorji
Slope correction: 1.093

No of clump	Clump condition*	Shoots/less than 1 year old culm	Bamboo stock			Harvestable culms **			Total no of culms
			1+ year (no)	2+years (no)	3+year (no)	1+ year (no)	2+years (no)	3+year (no)	
1	Good	5	7	8	8	6	7	7	20
2	Good	7	7	6	5	7	5	5	17
3	Congested	17	11	7	11	9	8	10	27
4	Congested	11	16	11	11	11	8	9	28
5	Congested	8	10	11	17	9	10	15	34
6	Congested	9	12	11	10	11	10	9	30
7	Good	4	6	5	6	5	4	6	15
8	Good	7	5	6	8	4	5	7	16
9	Congested	10	12	9	12	11	8	11	30
10	Damaged	3	3	1	1	2	1	1	4
11	Good	6	7	6	9	6	5	7	18
		87	96	81	98	81	71	87	239

Choeckhar Block

Date: 27/1/2024

Village: Ngangla

Recorder: Sha Kinley Doya & Kiran Mongar

Block: Choekar

Local name of resource assessment area: Choekar

Sample plot no: 2

Slope: 58%

Slope correction: 1.156

Aspect: NE

Altitude: 1395m

No of clump	Clump condition*	Shoots/less than 1 year old culm	Bamboo stock			Harvestable culms **			Total no of culms
			1+ year (no)	2+years (no)	3+year (no)	1+ year (no)	2+years (no)	3+year (no)	
1	Good	6	8	7	8	7	5	7	19
2	Good	7	5	7	6	5	6	5	16
3	Damaged	2	2	1	2	1	1	1	3
4	Congested	12	13	11	10	13	11	9	33
5	Good	3	5	5	7	4	4	5	13
6	Good	5	6	4	7	4	3	7	14
7	Damaged	1	3	0	2	3	0	2	5
8	Congested	12	14	13	12	13	12	11	36
9	Good	5	6	6	8	5	6	7	18
		53	62	54	62	55	48	54	157

Bangsekhar Block

Date: 29/1/2024

Village: Ngangla

Recorder: Sherab Dorji & Ugyen Dorji

Block: Bangsekhar

Local name of resource assessment area: Bangsekhar

Sample plot no: 3

Slope: Less than 45%

Slope correction: 1.093

Aspect: NW

Altitude: 1014m

No of clump	Clump condition*	Shoots/less than 1 year old culm	Bamboo stock			Harvestable culms **			Total no of culms
			1+ year (no)	2+years (no)	3+year (no)	1+ year (no)	2+years (no)	3+year (no)	
1	Good	7	6	5	7	5	4	6	15
2	Good	7	8	6	6	6	5	5	16
3	Congested	11	10	11	9	9	10	7	26
4	Good	6	4	4	7	4	3	5	12
5	Good	5	5	6	5	5	5	4	14
6	Congested	9	11	10	9	9	9	8	26
7	Good	3	5	6	7	5	5	4	14

8	Good	5	6	4	6	5	5	4	14
		53	55	52	56	48	46	43	137

Pongchaling Block

Date: 25/1/2024 **Village:** Pongchaling **Recorder:** Sha Kinley Doya & Kiran Mongar
Block: Pongchaling **Local name of resource assessment area:** Pongchaling
Sample plot no: 4 **Slope:** 55% **Slope correction:** 1.141
Aspect: NE **Altitude:** 1046m

No of clump	Clump condition*	Shoots/less than 1 year old culm	Bamboo stock			Harvestable culms **			Total no of culms
			1+ year (no)	2+years (no)	3+year (no)	1+ year (no)	2+years (no)	3+year (no)	
1	Good	4	5	6	7	5	5	4	14
2	Good	7	6	7	5	4	5	3	12
3	Good	8	5	5	7	3	3	6	12
4	Congested	12	8	7	8	6	6	7	19
5	Damaged	2	3	1	4	2	0	1	3
6	Damaged	1	1	2	2	0	0	1	1
7	Good	4	4	6	4	3	2	4	9
8	Good	5	4	6	8	4	4	6	14
9	Damaged	1	3	1	0	2	0	0	2
10	Good	6	5	4	4	5	3	2	10
		50	44	45	49	34	28	34	96

Appendix IV: Annual NWFP reporting form

Dzongkhag/Dungkhag:
Name of NWFP management and marketing group:
Gewog:
Villages:
Total number of Households:

NWFP Species	Quantity collected in a year	Qty sold in a year	Price received by the group per unit	Total income (Nu)	Description on resource availability	Remarks

Appendix V: List of NWFP Group Members

Sl. No.	Name	CID No.	Thram No.	House No.	Village
1	Rinchen Drakpa	12005001019	97	Wa-5-58	Sarimgang
2	Rinchen	12005001111	72	Wa-5-65	Ngangla Trong
3	Choizom	12005001215	1250	Wa-5-326	Ngangla Trong
4	Leki Chozom	11810001272	146	Wa-5-58	Ngangla Trong
5	Karma Chogyal	12005001255	22	Wa-5-75	Ngangla Trong
6	Tshewang	12005001233	28	Wa-5-74	Ngangla Trong
7	Nidup	12005001239	28	Wa-5-74	Ngangla Trong
8	Dendup	12005003029	855	Wa-5-286	Pongchaling
9	Choda	12005001774	18	Wa-5-10	Chekar
10	Pema Zangmo	12005001777	18	Wa-5-132	Chekar
11	Thinley	12005001752	17	Wa-5-09	Chekar
12	Pema Gyalpo	12005001187	24	Wa-5-70	Ngangla Trong
13	Pema Tshewang	12005001775	74	Wa-5-Nil	Ngangla Trong
14	Tshering Wangchuk	12005001180	64	Wa-5-69	Ngangla-Trong
15	Rinchen	12005001048	132	Wa-5-61	Ngangla Trong
16	Chogyal	12005001168	144	Wa-5-68	Ngangla Trong
17	Sonam Dorji	12005001002	206	Wa-5-57	Ngnagla Trong
18	Dorji Kencho	12005001673	175	Wa-5-04	Pongchaling
19	Tshewang	12005001033	101	Wa-5-59	Ngangla Trong
20	Mindu	12005001072	39	Wa-5-62	Ngangla Trong
21	Tshering	12005001041	102	Wa-5-60	Ngangla Trong
22	Rinchen Dorji	12005001201	7	Wa-5-71	Ngangla Trong
23	Karma Choda	12005001106	70	Wa-5-64	Ngangla Trong
24	Tempa	12005001703	306	Wa-5-06	Pongchaling
25	Mindu	12005001838	176	Wa-5-14	Pongchaling
26	Pelden	12005001057	720	Wa-5-46	Ngangla Trong
27	Gyalpo	12005001851	215	Wa-5-15	Pongchaling
28	Tashi Tshering	12005001225	425	Wa-5-18 Nil	Ngangla trong
29	Leki	12005001127	71	Wa-5-62	Ngangla Trong
30	Kinley Tshering	12005001128	430	Wa-5-66	Ngangla Trong
31	Sonam	12005001086	140	Wa-5-63	Ngangla Trong
32	Leki Dorji	12005001725	19	Wa-5-07	Chekar
33	Sonam Tobgay	12005001214	27	Wa-5-72	Ngangla Trong
34	Nima Dendup	12005001051	758	Wa-5-57 Nil	Ngangla Trong
35	Lhendup	12005001280	268	Wa-5-76	Ngangla Trong
36	Singye Dorji	12005000960	61	Wa-5-55	Ngangla Trong
37	Dendup	12005000980	206	Wa-5-57	Ngangla Trong
38	Lham Penjor	12005001115	145	Wa-5-67	Ngangla Trong
39	Dechen Yeshi	12005001222	23	Wa-5-73	Docheling
40	Sangay Pindu	12005001690	148	Wa-5-05	Pongchaling
41	Rigzen	12005001789	113	Wa-5-11	Pongchaling